Latte-Lab

A miniature conveyor belt system demonstrating the future of shopfloor automation. This project showcases a full-scale IIoT (Industrial Internet of Things) pipeline: from ESP32-driven physical motor control to real-time data visualization within SAP via MQTT and OPC-UA

Final Product

Convention

LARK & STERN
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Enable.
Efficient.
Delivery.

A13
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A13
BIOCAIR
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CORE SERVICES
Enable.
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AstraZeneca
AstraZeneca
Fasenra
Vannair
Symbicord
forxiga
Duforzig®
Dapagliflozin 10 mg
AstraZeneca

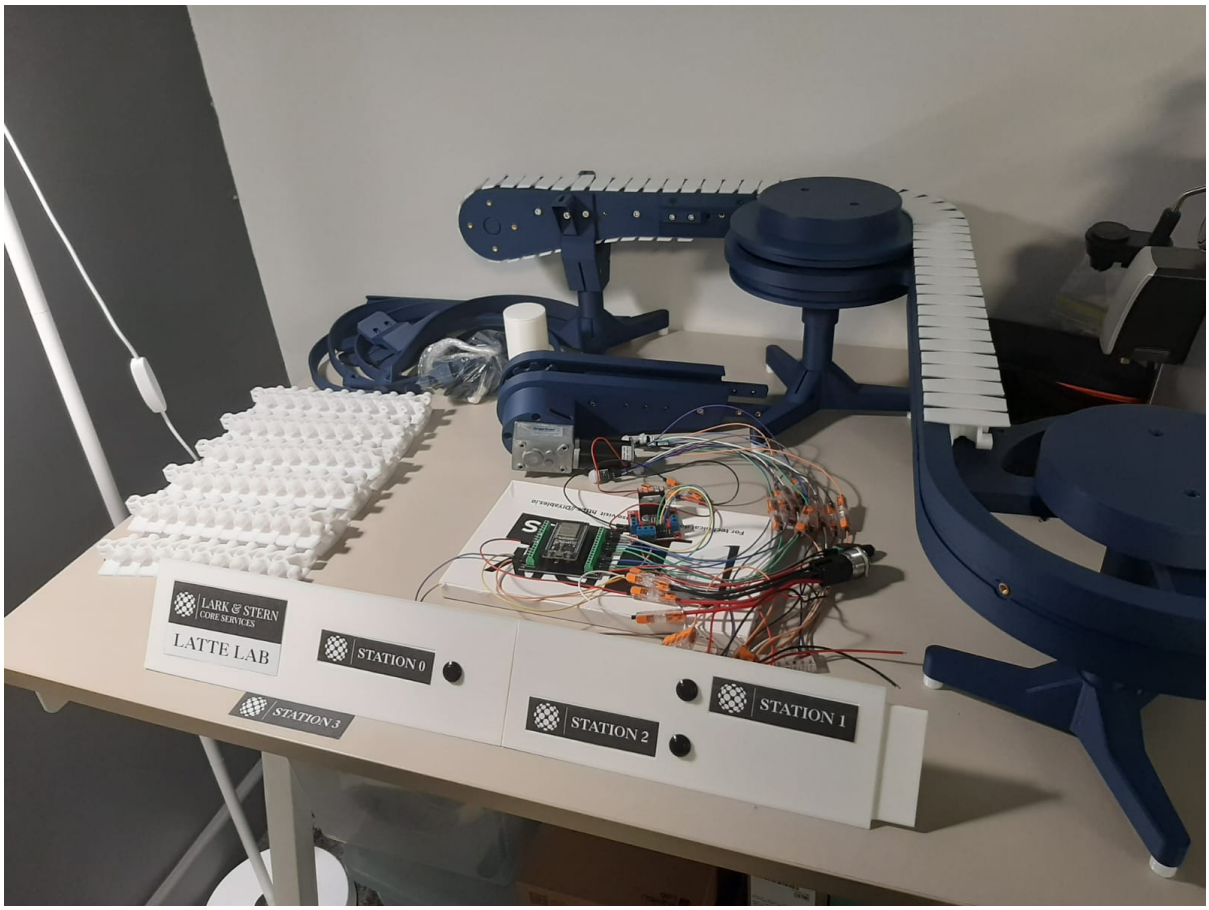
APEX SCIENTIFIC (PTY) LTD
VAISALA
Continuous Monitoring Systems
APEX SCIENTIFIC

SABINSA
BCX
STARRATE
NEW PRODUCT

Testo South Africa
A3
SABINSA

Other Visuals

Compass
RELOCATION
MEETUP2



Installation Manual

- Full Installation Manual

Wiring & Grounding Tips

Correct grounding is critical to prevent ESP32 "burnouts" or sensor interference.

> **Crucial Note:** Ensure the ESP32 Ground and the Motor Power Supply Ground are connected (Common Ground).

Wire Color Standards used in Latte Lab:

Type	Colors (In order of priority)
Ground	Black, Brown, Purple
VCC (3.3V)	Red, Orange, White/Gray
Functions	Yellow, Blue, Green

Note: Button wiring is pre-configured where Red is Power and Black is Ground.

Master Push Button Wiring

Button Pin Name	Position on Button	COnnected to	Wire Colour
IN NO1	(Top left)	Connect to the 3.3V pin	Red -> White
NO NO1	(Bottom left)	Connect to ESP32 Pin D25	Red -> Blue
IN NO2	(Top right)	Connect to ESP32 Pin D13	Black -> Yellow
NO NO2	(Bottom Right)	Not Connected	Black -> Nothing

Note: The "Position on Button" is from the perspective of looking at the button

Resource Links

- t0nyz - Bambu Conveyor for EPS32
- t0nyz - github.com/t0nyz0/Bambu-Poop-Conveyor-ESP32
- t0nyz - makerworld Bambu Poop Conveyor
- Modular Conveyor System
- Chain assembly instruction
- Chain installation instruction